



VISCOSITY



Let us consider a liquid, flowing in streamline motion, through a capillary tube, shown magnified in the fig. We can imagine the liquid in the tube to be made up of large number of co-axial cylindrical layers. It is found that the velocity of flow of all those layers are different, the velocity of the layer along the axis of the tube is maximum and it decreases as we move away from the axis and is zero for the layer in contact with the wall of the cylinder. This decrease in velocity of the liquid layers can be explained as due to a force, acting parallel to the surface of the liquid layers and opposite to the direction of motion. This hypothetical force frictional in nature is known as viscous force and the phenomenon is known as Viscosity.

Newton's formula for viscosity:

Let us consider a liquid layer in the tube of liquid

A = Surface area of that cylindrical liquid layer

$\frac{dv}{dr}$ = velocity gradient of that liquid layer

= Rate of change of velocity with distance.

Let F = Viscous force, acting on the liquid layer



According to Newton's law of viscosity

$$F \propto A \quad \text{and} \quad F \propto \frac{du}{dx}$$

$$\therefore F \propto A \frac{du}{dx}$$

$$\boxed{F = \eta A \frac{du}{dx}} \quad \text{--- (1)}$$

Where η = constant of proportionality and is known as Coefficient of viscosity.

If $A = 1$, $\frac{du}{dx} = 1$ then $\boxed{F = \eta}$

Definition: Thus Co-efficient of viscosity of a liquid can be defined as the viscous force, acting on unit surface area of that liquid layer at unit velocity gradient.

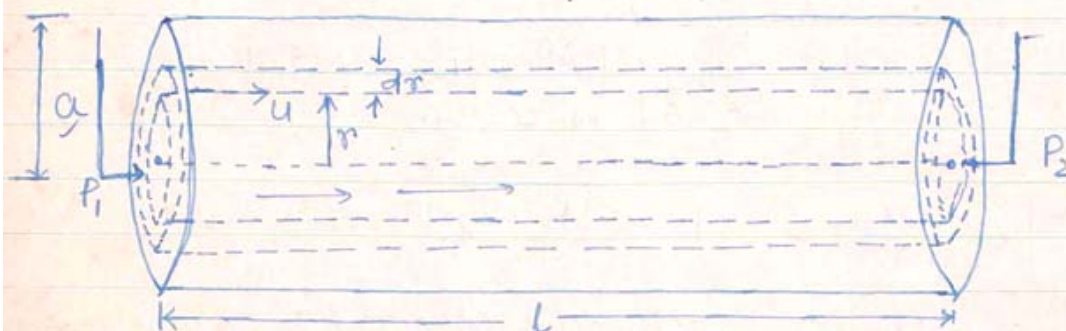
$$\therefore [\eta] = \frac{[F][dx]}{[A][du]} = \frac{[MLT^{-2}][L]}{[L^2][LT^{-1}]} = [ML^{-1}T^{-1}]$$

In C.G.S system, its unit is $\text{gm-cm}^{-1}\text{-sec}^{-1}$ (Poise)
In S.I system, its unit is $\text{kg-m}^{-1}\text{-sec}^{-1}$.



Flow of liquid through a Capillary tube :-
Poiseuille's formula :-

Let us consider a liquid flowing in streamline motion through a capillary tube, shown magnified in the figure.



Given : l = The length of the capillary tube.
 a = The radius of the capillary tube.
 P = The pressure difference across the capillary tube = P_{inlet} at the inlet (P_1) - P_{outlet} at the outlet.
 η = Co-efficient of viscosity of the liquid.

We can imagine the liquid flowing through the tube is made up of, large number of co-axial cylindrical layers. Let us consider one such cylindrical liquid layer of radius ' r ' ($0 \leq r < a$)

Let u = velocity of flow of that liquid layer.
 and $\frac{du}{dr}$ = velocity gradient at that liquid layer.

Using Newton's law of viscosity, the viscous force acting on the liquid layer,

$$F = \eta A \cdot \left(-\frac{du}{dr}\right)$$

The -ve sign indicates that velocity decreases as the



Or, $F = -\eta(2\pi r l) \cdot \frac{du}{dr}$ from right to left, opposite to the direction of motion flow.

The force acting on this liquid cylinder from left to right, due to the pressure difference.

$$F' = P_1 \pi r^2 - P_2 \pi r^2 = (P_1 - P_2) \pi r^2 = P \pi r^2$$

Since we observe the motion at steady state; the velocity of a given liquid layer remains constant with respect to time, $\frac{du}{dt} = 0$, hence the motion is unaccelerated. So the resultant force on that liquid layer must be zero i.e. force towards right due to the pressure difference is equal to viscous force towards the left.

$$P \pi r^2 = -2\pi \eta l \cdot \frac{du}{dr}$$

$$\therefore \boxed{\frac{du}{dr} = -\frac{P \cdot r}{2\eta l}} \quad \text{--- (1)}$$

From equation (1) we find that, the velocity gradient depends on the radius of the liquid layer and increases with the increase in radius of the layer.

Equation (1) can be rearranged as

$$du = -\frac{P}{2\eta l} r dr$$

Integrating on both the sides

$$\int_0^u du = \int_a^r -\frac{P}{2\eta l} r dr$$

$$\therefore u = -\frac{P}{2\eta l} \left[\frac{r^2}{2} - \frac{a^2}{2} \right]$$



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$$P \pi r^2 = -2\pi \eta l \cdot \frac{du}{dr}$$

$$\therefore \left[\frac{du}{dr} = -\frac{P \cdot r}{2\eta l} \right] \quad (1)$$

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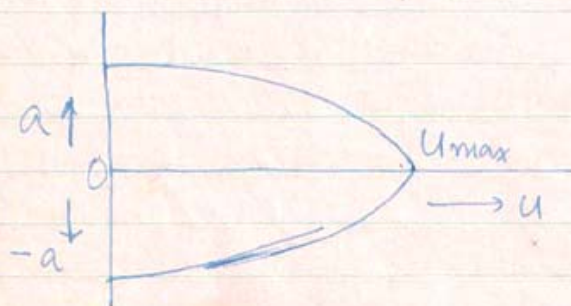
$$\therefore u = -\frac{P}{2\eta l} \left[\frac{r^2}{2} - \frac{a^2}{2} \right]$$



$$\therefore u = \frac{P}{2\eta l} \frac{(a^2 - r^2)}{2}$$

$$\therefore \boxed{u = \frac{P}{4\eta l} (a^2 - r^2)} \quad \text{--- (2)}$$

From equation (2) we can find that the velocity of different liquid layers, of different radius are different. The variation of the velocity of the layers with changing radius is shown below.



We now find the volume of liquid flowing through the capillary tube per second -

Since the velocity of all the liquid layers are different; let us imagine, two co-axial cylinders of radii ' r ' and ' $r + dr$ ' where $0 < r < a$. The annular gap ' dr ' between the two cylinders is so small that velocity of all the liquid layers in the gap can be assumed to be same, equal to u .



Area of cross-section of the annular space = Circumference $\times$ thickness
 $= 2\pi r dr$

Hence the volume of the liquid flowing through the annular gap between the two co-axial cylinders per second dV = area of cross-section $\times$ distance covered in 1 sec (velocity)



$$\text{or } dv = 2\pi r dr \cdot u$$

Putting equation (2):

$$dv = \frac{2\pi r dr \cdot P (a^2 - r^2)}{4\eta l}$$

$$dv = \frac{P\pi}{2\eta l} [a^2 r dr - r^3 dr] \text{ --- (3)}$$

Hence the total volume of liquid flowing through the capillary tube in 1 second, can be obtained by integrating equation (3):

$$V = \int dv = \int_0^a \frac{P\pi}{2\eta l} [a^2 r dr - r^3 dr]$$

$$V = \frac{P\pi}{2\eta l} \left[\int_0^a a^2 r dr - \int_0^a r^3 dr \right]$$

$$V = \frac{P\pi}{2\eta l} \left[a^2 \cdot \frac{a^2}{2} - \frac{a^4}{4} \right] = \frac{P\pi a^4}{8\eta l}$$

$$V = \frac{P\pi a^4}{8\eta l} \text{ --- (4) or } \eta = \frac{P\pi a^4}{8Vl} \text{ --- (5)}$$

Equation (4) which gives the volume of liquid flowing per second through the capillary tube is known as Poiseuille's formula.

Correction: In deriving equation (4), the following assumptions have been made

- (a) There is no acceleration, along the axis of the tube.
- (b) There is no kinetic energy generated.
- (c) All the work done by the pressure difference is converted into heat by the viscous force.



(c) It is further assumed that the applied pressure difference is totally used in overcoming viscous resistance. But in actual practice a part of the applied pressure is used to give K.E to the liquid

between the liquid layers.

However none of the above assumptions are quite correct.

(a) Along the axis of the tube, there is always some acceleration, just near the inlet and outlet of the tube. This can be corrected by using $[l + 1.64a]$ instead of 'l' in equation (4).

(b) Correction due to K.E:

work done due to the pressure difference per second = work done by the viscous force per second + K.E generated per second — (5)

work done per second by the pressure

$$W = F \times S$$

S = dist. per second = velocity

$$= P (\text{area of cross-section} \times u)$$

$$= PV$$

V = volume flowing per second through the capillary tube

Using equation (6) we can write :-

$$PV = P_1 V + \text{K.E generated per second} \text{ — (6)}$$

where P_1 = effective pressure difference which has to overcome the viscous force.

Hence equation (2) and (4) can be written as

$$u = \frac{P_1 (a^2 - r^2)}{4\eta l} \text{ — (2a)}$$

$$V = \frac{P_1 \pi a^4}{8\eta l} \text{ — (4a)}$$



Let us consider the liquid flowing through the annular gap between the two ~~co-axial~~ co-axial cylinders of radius r & $r+dr$.

The K.E generated per second for that liquid
 $= \frac{1}{2} \left(\text{mass of the liquid flowing per second through the annular zone} \right) \times u^2$

$$= \frac{1}{2} \left[\rho (2\pi r dr \times u) \right] u^2$$

$$= \pi \rho r dr u^3$$

Putting equation (2a) $= \frac{\pi \rho r dr (a^2 - r^2)^3}{(4\eta l)^3} \rho_1^3$

The total K.E generated per second.

$$= \int_0^a \left(\frac{\rho_1}{4\eta l} \right)^3 (a^2 - r^2)^3 r dr \quad \text{--- (8)}$$

$$\Rightarrow \int_0^a (a^2 - r^2)^3 r dr = \int_0^a [a^6 - 3a^4 r^2 + 3a^2 r^4 - r^6] r dr$$

$$= \left[\int_0^a a^6 r dr - 3a^4 \int_0^a r^3 dr + 3a^2 \int_0^a r^5 dr - \int_0^a r^7 dr \right]$$

$$= \frac{a^8}{2} - \frac{3a^8}{4} + \frac{3a^8}{6} - \frac{a^8}{8} = a^8 \left[\frac{12 - 18 + 12}{24} \right]$$

$$= \frac{a^8}{8}$$

$\therefore$ The K.E of the liquid generated per second:

$$= \left(\frac{\rho_1}{4\eta l} \right)^3 \frac{\pi \rho a^8}{8} = \left(\frac{\rho_1}{4\eta l} \right)^3 \left(\frac{\pi a^4}{2} \right) \frac{\rho}{\pi^2} \cdot \frac{1}{a^4}$$

$$= \left(\frac{\rho_1 \pi a^4}{8\eta l} \right)^3 \frac{\rho}{\pi^2 a^4}$$



Putting Equation (4a) :

$$\text{K.E generated per second} = \frac{\rho v^3}{\pi^2 a^4} \quad \text{--- (9)}$$

Putting equation (9) in (7)

$$Pv = P_1 V + \frac{\rho v^3}{\pi^2 a^4}$$

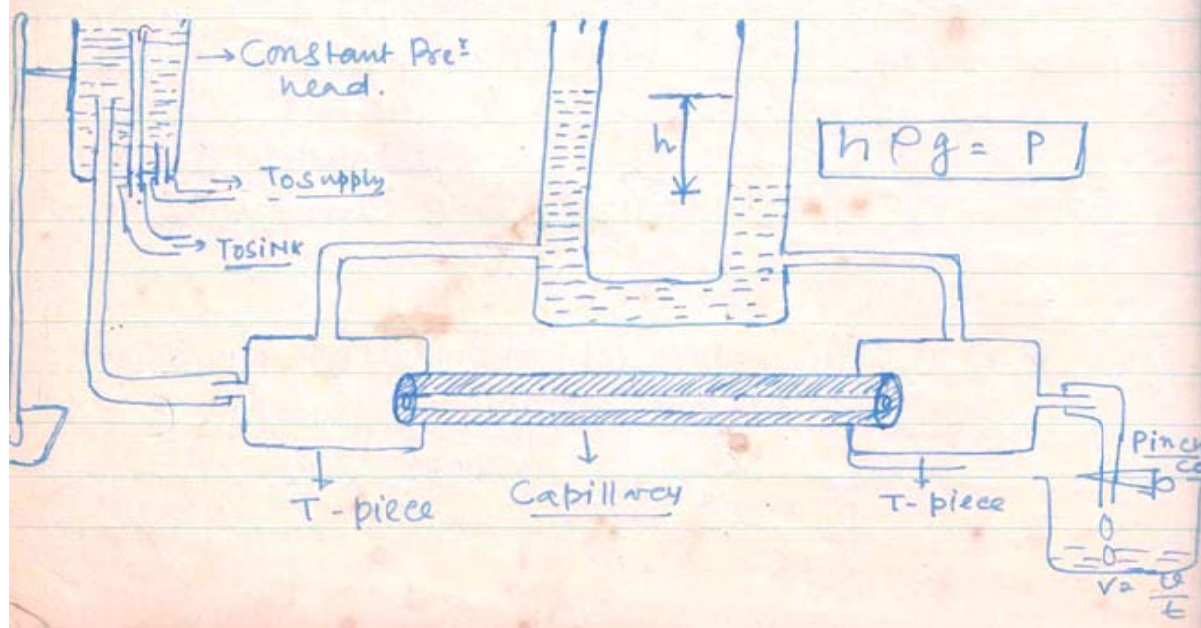
$$P_1 = P - \frac{\rho v^2}{\pi^2 a^4} \quad \text{--- (10)}$$

Hence the corrected form of equation (5) :-

$$\eta = \frac{\pi a^4 [P - \frac{\rho v^2}{\pi^2 a^4}]}{8v [L + 1.64a]} \quad \text{--- (11)}$$

Using equation (11) coefficient of viscosity of a liquid can be experimentally determined.

Experimental arrangement :





Experimental liquid from a constant pressure head is made to flow through a horizontal capillary tube, having two T-pieces connected at its two ends. The two ends of the capillary tube are connected to the two limbs of a 'U' tube, where the liquid rises up to different heights, depending on the pressure at the inlet and outlet of the tube. The difference in liquid levels in a 'U'-tube should be kept constant at 8 to 12 cm by controlling the rate of outflowing liquid by the pinch cock & the outflowing liquid is allowed to fall by trickles & is collected in a beaker for a time interval recorded by a stop-clock. The volume of collected liquid is measured. The difference in level is also recorded. The radius of capillary tube is measured by a travelling microscope.

Calculation:-

h = difference in liquid levels in U-tube.

ρ = density of the liquid.

V = Volume of the outflowing liquid collected in t sec.

$\therefore$ The pressure difference across the capillary tube
 $P = h\rho g$.

The volume of the liquid flowing per second through the capillary tube $V = \frac{V}{t}$

Knowing P, V, a & L ; η can be calculated using equation (1).